Writabrata Bhattacharya

Applied Scientist | GenAI · LLMs · Speech AI · MLOps · Time-Series Forecasting

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Experience

Inception Institute of Artificial Intelligence, Applied Scientist · Machine-Readable Government (DGE Abu Dhabi) — Abu Dhabi, UAE — July 2024 – present — 2 years 4 months

- Built a production **LLM compliance-automation platform** for government product-registration review, turning **policy PDFs into machine-readable policy graphs** and automating case-by-case compliance assessments
- Architected a **4-stage multi-agent workflow** (retrieval → taxonomy → compliance → decision) on the **OpenAI Agents SDK**, adjudicating cases with **sandboxed, tool-using agents** and rule-guided decisions
- Engineered **hybrid retrieval** — **OpenAI Vector Store** semantic search for recall, local **policy-graph validation** for precision — for deterministic, cross-referenced policy expansion per case
- Built a **PDF → policy-graph extraction pipeline** with **3-stage LLM validation**, preserving clause locators and resolving cross-policy replaces / cancels / amends links for auditable checks
- Developed an **evaluation framework** across retrieval, compliance, decision, and graph accuracy, run through a **containerized, sharded, resumable harness**, with **Langfuse + OpenTelemetry** observability
- Deployed on **Azure Kubernetes** with **Terraform IaC** and Service Bus–throttled workers; secured with **Entra ID (OAuth2 + PKCE)**, **Postgres row-level multi-tenancy**, and a **Ruff / pytest / Checkov** CI/CD gate

Inception Institute of Artificial Intelligence, Applied Scientist · EnergyAI (ADNOC) — Abu Dhabi, UAE

- Led design and development of advanced **RAG systems** for upstream **oil & gas** operations, handling structured and unstructured data to deliver context-aware responses
- Optimized the RAG pipeline with **Azure Document Intelligence** for PDF parsing and **dynamic semantic chunking**, minimising context loss
- **Fine-tuned Meta's LLaMA 70B** (PEFT and full fine-tuning) and **LoRA-tuned Qwen3-Embedding-8B** with **model merging** (arXiv:2601.04720) for superior domain-specific retrieval
- Built an end-to-end **evaluation framework** auditing every RAG stage — OCR, chunking, retrieval accuracy, generation fidelity — for systematic performance tracking
- **Benchmarked** state-of-the-art and open-source **LLMs and embedding models** on oil & gas domain data
- Built a **Chatbot Arena-style evaluation frontend** for direct LLM comparison and ran **load / stress testing** for production deployment

Boston Consulting Group, Data Scientist Intern — Mumbai, India — Oct 2023 – Apr 2024 — 7 months

- Increased call **transcription accuracy** by analysing audio to identify language and tailor transcription strategies
- Built a complete **audio-processing pipeline** — **VAD** noise reduction, **Mel-Spectrogram** features, and a **TDNN language-identification** model
- **Fine-tuned Whisper** with hyper-parameter tuning for Indian languages, significantly improving transcription fidelity
- Engineered a **QA module** to flag recurring lines and timestamp discrepancies; used **GPT-3.5 and XLM-RoBERTa** for business-category classification

Skills

Machine Learning & AI: LLMs, RAG, Fine-tuning (LoRA/PEFT), TTS, STT, NLP, Embeddings, Transformers, Reinforcement Learning

MLOps & Engineering: Docker, Kubernetes, FastAPI, Azure, Databricks, vLLM, Gradio, OpenTelemetry, Postgres/SQLModel, Load Testing, Production Deployment

Statistics & Mathematics: Time Series, Stochastic Processes, Bayesian Inference, Measure Theory, Statistical Learning

Programming: Python, R, Haskell, C++, JavaScript

Frameworks & Tools: PyTorch, LangChain, Hugging Face, OpenAI Agents SDK, Langfuse, verl, vLLM

Education

PGDBA — **Indian Institute of Technology Kharagpur**, Data Science — Kharagpur, India — Aug 2022 – May 2024

- Relevant Coursework: Statistical Learning, Deep Learning, Bayesian Inference, Categorical Data Analysis
- Graduated in the **top 10%** of the cohort

MSc — **Chennai Mathematical Institute**, Pure Mathematics — Chennai, India — Aug 2020 – June 2022

- Relevant Coursework: Measure Theory, Stochastic Process, Operator Theory, Ergodic Theory
- **Thesis:** Angle & Distance Estimates and Kazhdan Constants
- Full academic scholarship and tuition fee waiver

BSc — **Chennai Mathematical Institute**, Mathematics and Theoretical Computer Science — Chennai, India — Aug 2017 – June 2020

- Relevant Coursework: Probability, Algebra, Topology, Manifolds, Riemannian Geometry, Algorithms
- Full academic scholarship and tuition fee waiver

Publications

PETRA: Transforming Web Text for Petroleum-Engineering Domain Adaptation — June 2026
K. Dubovikov, O. El Mansouri, H. Madmoun, Y. Li, S. Kumar, A. El Mir, S. Ghosh, **W. Bhattacharya**, A. Garcia-Garcia, O. Pandit, S. K. Sahu, F. Castanedo, L. Murray, M. Takac, S. Lahlou
arxiv.org/abs/2606.24346 (arXiv preprint (arXiv:2606.24346))

Assessing The Capabilities Of Large Language Models For Oil And Gas Industry Applications — Nov 2024
W. Bhattacharya, et al.
onepetro.org/SPEADIP/proceedings/24ADIP/24ADIP/D021S033R003/579034 (SPE Journal (SPE-229701-MS) — presented at ADIPEC 2024)

Awards

Olympiads – INMO, INPhO, INAO: Selected among India's top 30 students in the Mathematics (INMO), Physics (INPhO), and Astronomy (INAO) Olympiads; advanced to the national training camp at HBCSE, TIFR.

INSPIRE Scholarship: Government of India – DST. Awarded for outstanding academic performance (₹1.8L p.a. for 3 years).

JEE Mains – AIR 1018: Top 0.085 percentile out of 1.2 million candidates (2017).

JEE Advanced – AIR 1039: Top 0.6 percentile in JEE Advanced (2017).

CMI Full Scholarship: Full academic scholarship and tuition fee waiver for both the B.Sc. and the M.Sc., awarded by Chennai Mathematical Institute.

Leadership

Editor-in-Chief – AINA 4.0: PGDBA Annual AI and Analytics Magazine. Led a team of 6; featured an interview with Dr. Josh Starmer (StatQuest).

Olympiad Mentor – Cheenta Ganit Kendra: Mentored Mathematics and Physics Olympiad aspirants for four years (2017–2021).

Languages

English: Fluent

Bengali: Native

Hindi: Fluent

Spanish: Beginner

Interests

Research: World Models, Reinforcement Learning, LLMs, Speech AI, Quantitative Finance, Game Development